

and code that acts on the data exist in one grouping. This also tends to make code easier to debug.

3. Objects created from a class only have access to their own data—their set of the instance variables in the class. Even when you have multiple objects created from the same class, they do not have access to each other's data.

Summary

In this chapter, I provided an introduction to object-oriented programming by demonstrating the relationship between a class and an object. The class defines the shape and capabilities of an object. An object is a single instance of a class that has its own set of all the data defined in the instance variables of the class. Each piece of data you want an object to contain is stored in an instance variable, which has object scope, meaning that it is available within all methods defined in the class. All objects created from the same class get their own set of all the instance variables, and because these may contain different values, calling the methods on different objects can result in different behavior.

I showed how you create an object from a class, typically through an assignment statement. After instantiating an object, you can use it to make calls to any method defined in the class of that object. I also showed how you can instantiate multiple objects from the same class.

In this chapter, the demonstration classes implemented physical objects (light switches, TVs). This is a good way to start understanding the concepts of a class and an object. However, in future chapters, I will introduce objects that do not represent physical objects.